

Watering Dashboard

Setup Guide & Documentation

A full-stack web application for managing watering zones and plants with interactive maps — built with **React + Leaflet** (frontend) and **Node.js + Express + MySQL** (backend).

Supports creating and editing areas (zones), placing plants/watering stations on maps, photo uploads, and an audit log for all user actions.

1. Requirements

Make sure the following are installed on your system before proceeding:

- **Node.js** (v18 or newer)
- **npm** (comes bundled with Node)
- **MySQL Server**
- **Git** (optional, for cloning the repository)

2. Backend & Database Setup

Step 1 - Install dependencies

```
npm install i
```

Step 2 - Create .env file

Create a file named `.env` in your root server directory and populate it with the following configuration details:

```
DB_HOST=localhost  
DB_USER=root  
DB_PASSWORD=your_mysql_password  
DB_NAME=watering_db  
PORT=3000  
NODE_ENV=development
```

Step 3 - Create the database in MySQL

Execute the following SQL command within your MySQL client to instantiate the application database:

```
CREATE DATABASE watering_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
```

Step 4 - Run database migrations

Navigate to the database migration folder and run the execution script matching your platform:

```
cd database
# On Windows
powershell -ExecutionPolicy Bypass -File run_migrations.ps1

# On Linux/Mac
bash run_migrations.sh
```

Step 5 - Start backend

Launch the Express server to handle API calls and connections:

```
cd server
npm install
node server.js
```

3. Frontend Setup (React)

Step 1 - Go to frontend folder

```
cd client
```

Step 2 - Start frontend

Run the application server depending on your build configuration:

```
npm start
```

If your project is scaffolded using Vite, use:

```
npm run dev
```

4. Environment Notes

- The `.env` configuration file must exist in the designated directory before running the backend service.
- By default, the **Backend** service runs on `http://localhost:3000`
- The **Frontend** web client boots on `http://localhost:5173` (if using Vite) or `http://localhost:3000` (if using Create React App).
- Cross-Origin Resource Sharing (**CORS**) is fully enabled for local development workflows.

5. Usage Flow

1. Register a new user account through the interface.
2. Login with your created credentials.
3. The system authenticated session opens the main dashboard.
4. Add or edit watering zones (areas) using the dynamic map interface.
5. Place individual plants or structural watering stations directly onto the interactive map layers.
6. Upload real-time photos for designated areas and plants.
7. Monitor and review the built-in audit logs for all ongoing system actions.